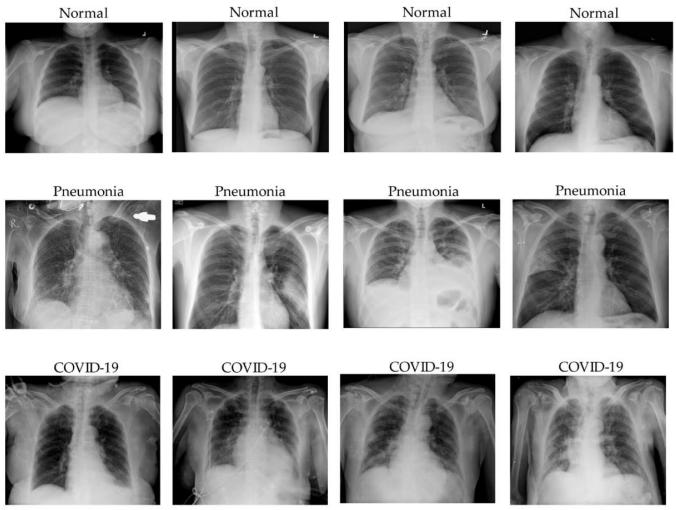
Convolutional Neural Networks

Ralf Gabriels

Machine learning methods in biomedical data
Academic year 2025-2026

There are various prediction tasks in computer vision:
Classification



Le Dinh, T, et al. COVID-19 Chest X-ray Classification and Severity Assessment Using Convolutional and Transformer Neural Networks. Appl. Sci. 2022, 12, 4861

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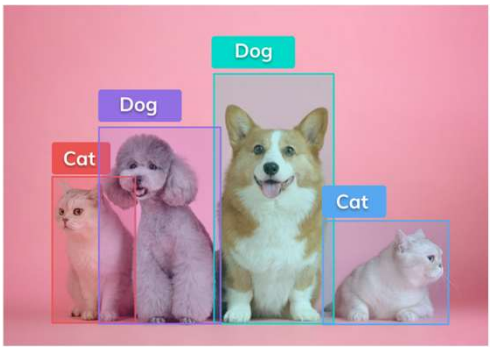
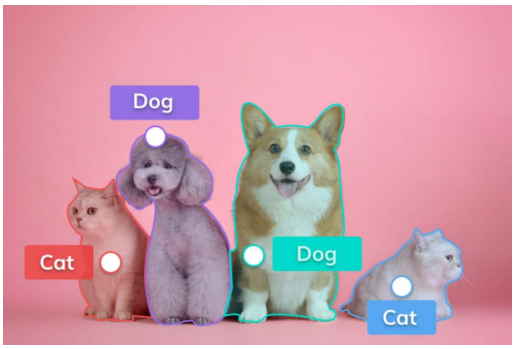
In computer vision, one important task is **image classification**, where the goal is to assign a single label to an entire image. In biomedical contexts, this could mean predicting whether an X-ray is normal or abnormal, or assigning an image to a disease category such as COVID-19 or pneumonia. The model does not indicate where the relevant pattern is located; it only predicts the overall class of the image.

There are various prediction tasks in computer vision:
Detection & segmentation





Le Dinh, T, et al. COVID-19 Chest X-ray Classification and Severity Assessment Using Convolutional and Transformer Neural Networks. Appl. Sci. 2022, 12, 4861

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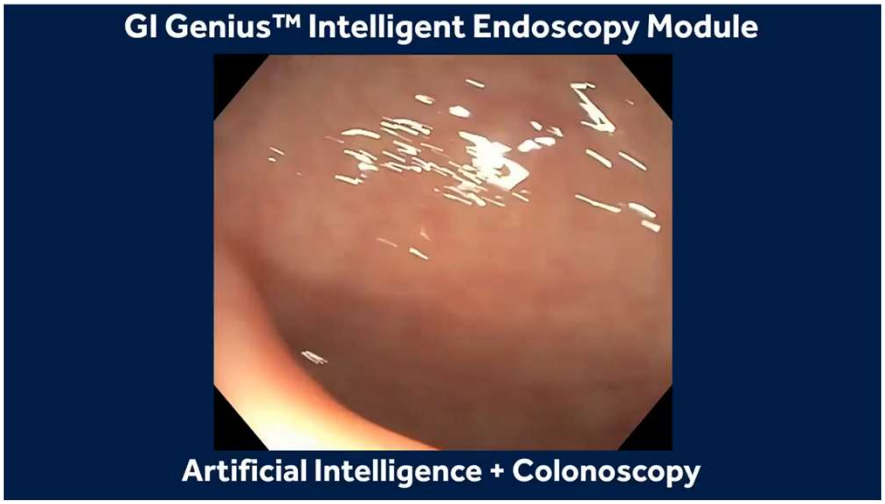
Computer vision tasks can also require more detailed outputs than a single class label. In **object detection**, the model identifies and localizes objects, typically with bounding boxes. In **segmentation**, the model predicts which individual pixels belong to which object or region. These tasks are especially important in biomedical imaging, where one may want to detect lesions, outline tumors, or separate cells from background tissue.

There are various prediction tasks in computer vision:
Object detection



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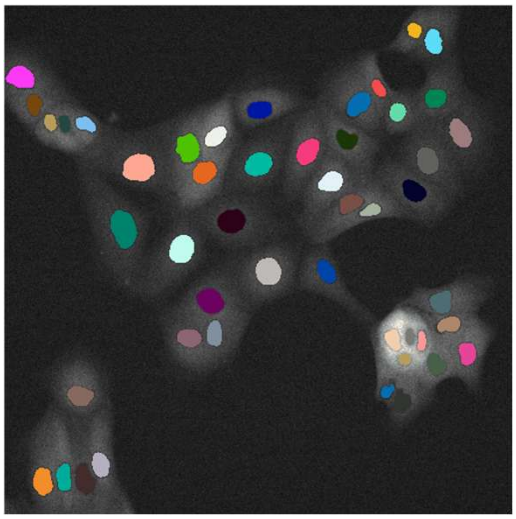
In **object detection**, determining where it is located can be useful in applications such as endoscopy, pathology, or radiology, where the location of a suspicious region matters as much as its presence. Detection therefore extends classification by adding spatial localization.

There are various prediction tasks in computer vision:
Object tracking



https://imaging.cs.msu.ru/en/research/cell-tracking

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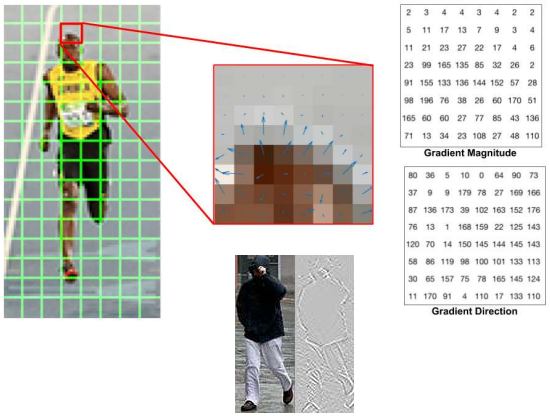
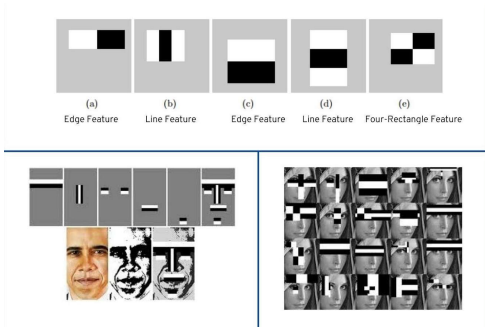
Some computer vision problems involve following objects over time rather than analyzing a single static image. In **object tracking**, the model identifies the same object across consecutive frames of a video or time-lapse experiment. In biomedical research, this is commonly used to track cells, organelles, or other structures as they move, divide, or interact over time.

Methods before CNNs involved manual feature extraction



Histogram of oriented gradients



Edge detection

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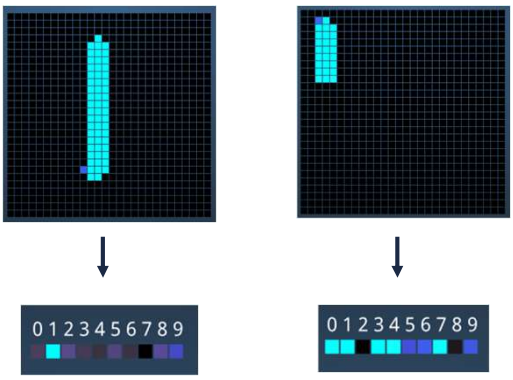
Before CNNs became dominant, image analysis often relied on **manual feature extraction**. Researchers designed features such as edges, gradients, textures, or shapes and then passed those features to a separate machine-learning model. This approach could work well, but it depended heavily on expert knowledge and on the assumption that the manually designed features captured the relevant biological or visual information.

What if neural networks could learn visual patterns instead?

Regular neural network



When digit is centered: good prediction

When digit is NOT centered: bad prediction

- ✅ Can learn visual patterns itself!
- ❌ Expects the image to stay in the same location (not robust against translation invariance).

https://adamharley.com/nn_vis/mlp/2d.html

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A standard fully connected neural network can, in principle, learn from image pixels directly. However, it treats the image as a collection of input values rather than as a structured 2D object. As a result, it is sensitive to the exact position of patterns in the image. If an object shifts to another location, the network may respond very differently, even though the image content is biologically or visually the same.

Image recognition must be robust against variances

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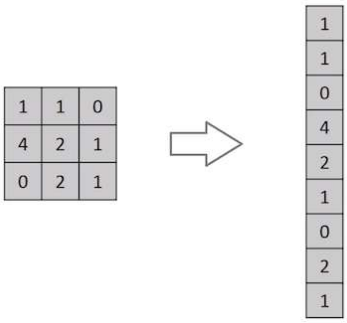
Real-world images vary in many ways. Objects can appear at different positions, sizes, orientations, or lighting conditions. These are called object **invariances**. A useful image-recognition model must therefore be robust to such variations. In biomedical imaging, similar challenges arise because samples may differ in staining intensity, imaging angle, magnification, or acquisition conditions. CNNs were designed in part to better handle this type of variability.

A regular neural network requires flattening of the 2D image, rendering the model not robust against translation invariance



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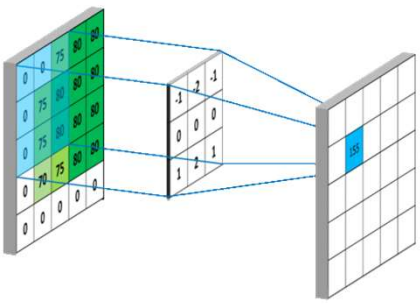
A regular neural network requires the image to be **flattened** into a one-dimensional vector before it can be processed. This conversion removes the explicit spatial arrangement of neighboring pixels. Consequently, the model no longer naturally knows which pixels were adjacent in the original image, making it harder to recognize the same pattern when it appears in a different location.

Kernels (aka filters) solve the problem of translation invariance

Kernels 'convolve' over the image



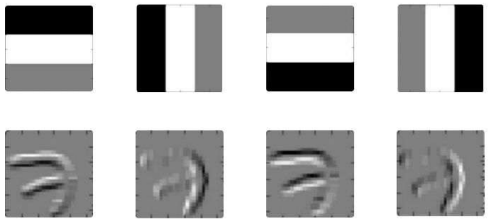
Have been used for manual feature engineering, for instance, for edge detection

-1	-1	-1
1	1	1
0	0	0

-1	1	0
-1	1	0
-1	1	0

0	0	0
1	1	1
-1	-1	-1

0	1	-1
0	1	-1
0	1	-1



Manual edge detection figure: https://octopusinvitro.gitlab.io/blog/machine-learning/convolutional-networks

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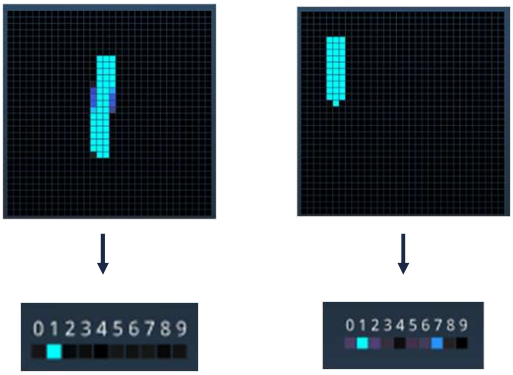
CNNs address this limitation by using **kernels**, also called **filters**, that slide across the image. A kernel applies the same small set of weights at many positions, allowing the model to detect the same local pattern wherever it appears. This idea is related to earlier image-processing methods such as edge-detection filters, but in CNNs the filters are learned from data rather than manually specified.

Kernels (aka filters) solve the problem of translation invariance

Convolutional neural network



Good prediction independently of object location

- ✅ Can learn visual patterns itself!
- ✅ Is robust against translation invariance

https://adamharley.com/nn_vis/cnn/2d.html

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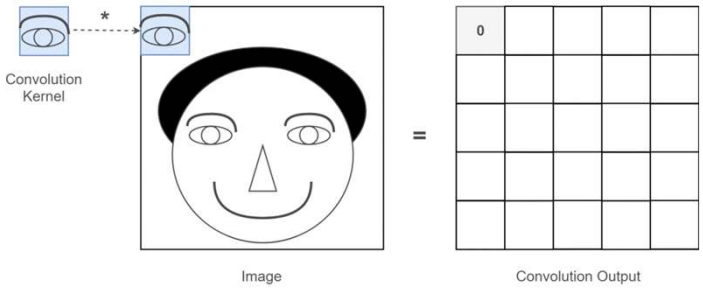
Because the same filter is applied across the entire image, a convolutional neural network can recognize a pattern even when it appears at different positions. This makes CNNs much more robust to **translation invariance** than standard fully connected networks. The model still learns from data, but it does so in a way that respects the **spatial structure** of images.

Kernels, simplified



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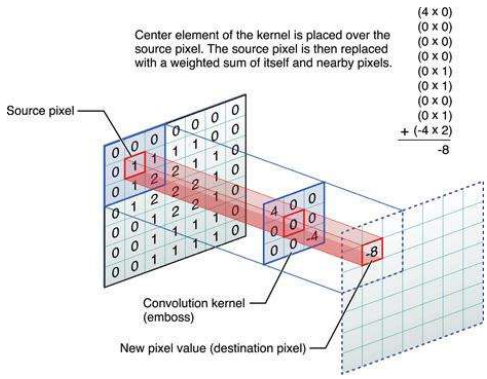
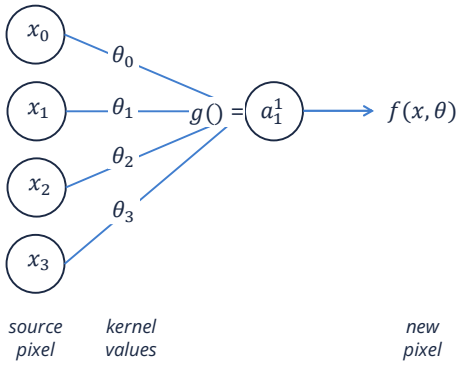
A kernel can be understood as a small window that scans over the image and computes a new value at each position. The output records how strongly the local image region matches the pattern encoded by the kernel. Repeating this operation over the whole image produces a **feature map**, which highlights where that visual pattern is present.

In CNNs, kernels values are learnable parameters of the model





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In a CNN, the values inside a kernel are not fixed by the user. They are **learnable parameters** that are optimized during training, just like weights in a dense neural-network layer. During learning, the model adjusts these filter values so that the resulting feature maps become useful for the prediction task. In this sense, a convolution is a local weighted combination of nearby pixels.

Multiple convolutions can be stacked in layers

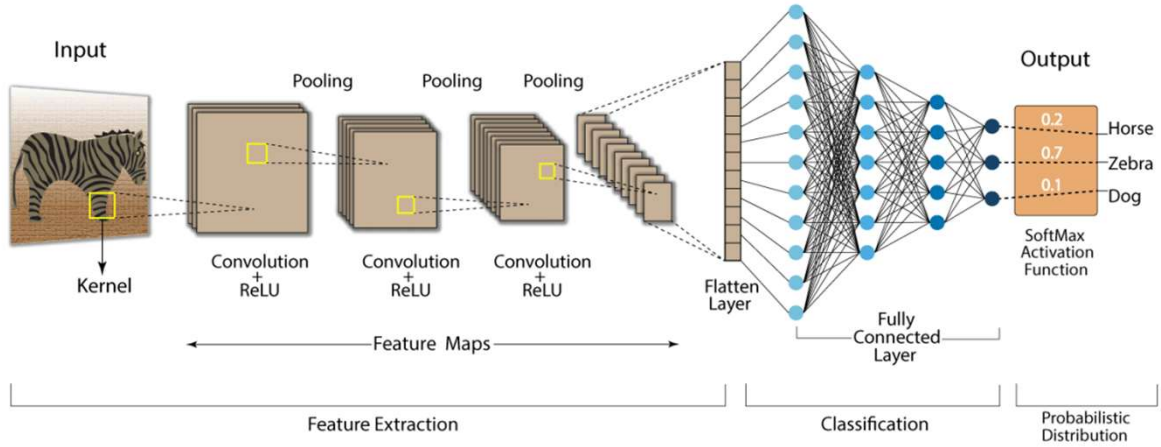

Image from developersbreach.com

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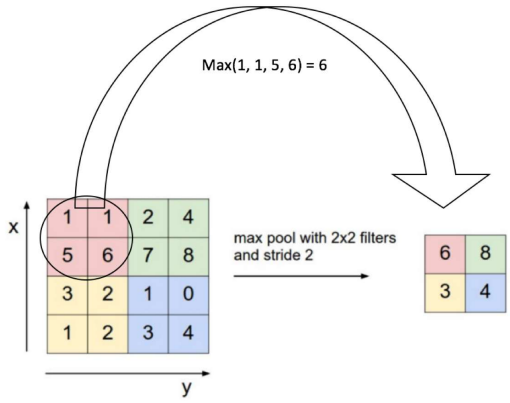
CNNs become powerful because multiple convolutional **layers can be stacked**. Early layers often detect simple patterns such as edges or textures. Later layers combine these simpler patterns into more complex structures. In biomedical imaging, this hierarchy might progress from local contrast changes to tissue architecture, cell morphology, or lesion patterns. The final layers then use these learned representations for classification or another downstream task.

Pooling operations downscale the feature map (or image)



original image

after max pooling

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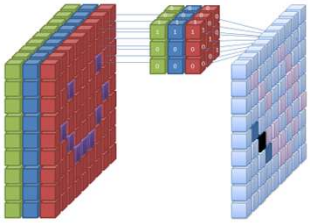
Pooling reduces the spatial size of a feature map by summarizing small local regions, for example by taking the maximum value. This makes the representation more compact and reduces computational cost. Pooling can also make the model less sensitive to small local shifts, because the exact pixel position becomes less important once nearby responses are aggregated.

Color images have multiple channels, but so have (some) kernels

Source images can have red, green, and blue color channels

Kernels have according channels too



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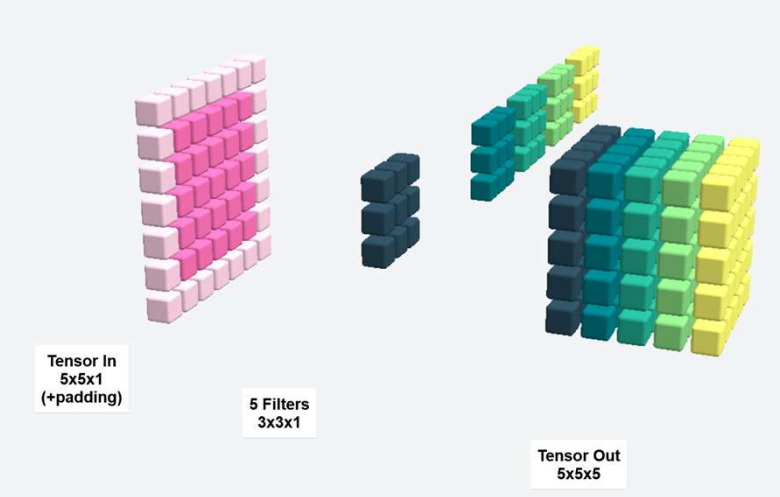
Many images contain multiple **channels**. In natural images these are typically red, green, and blue, while in biomedical imaging they may correspond to grayscale intensity, fluorescence channels, stains, or imaging modalities. Convolutional kernels must match this channel structure: a filter operating on a multi-channel image learns weights across all channels simultaneously, allowing it to combine information from them into a single response.

Applying multiple kernels over the same image also leads to multi-channel 'images' (or feature maps)



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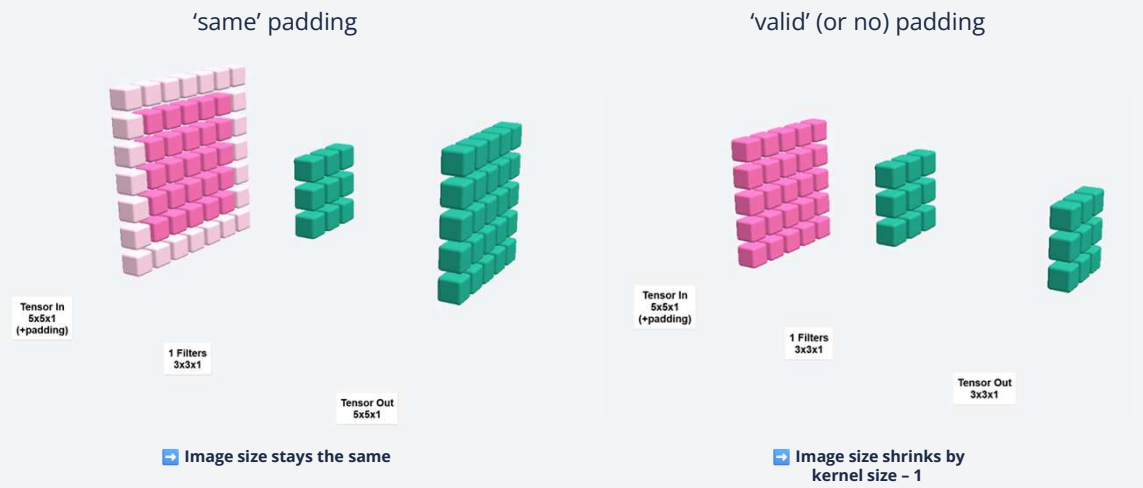
A single convolutional layer usually applies **many different kernels** to the same input image or feature map. Each kernel learns to detect a different type of pattern, and each one produces its own output channel. The result is therefore not a single image, but a stack of newly generated images (also called feature maps). This allows the network to represent many visual characteristics at the same layer.

Padding determines the shape of the output image

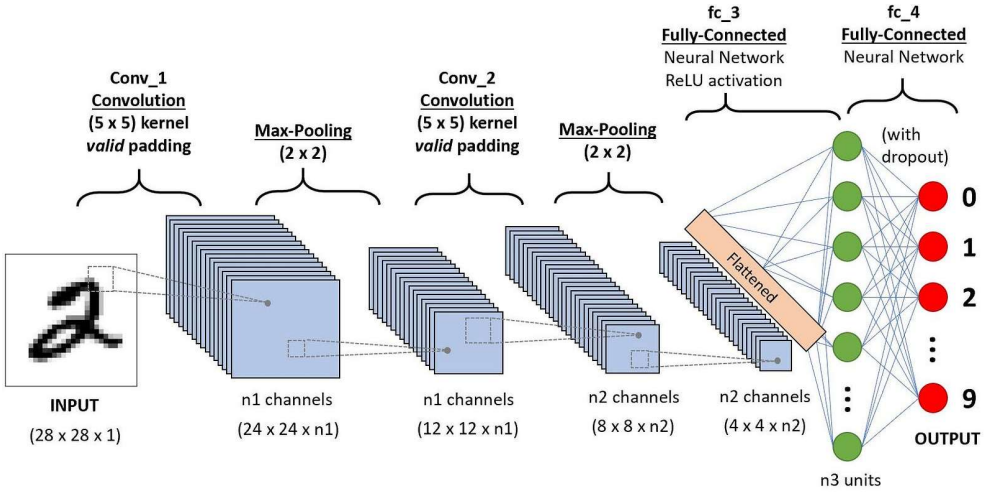


18

When a kernel is applied near the border of an image, there may not be enough surrounding pixels to fully cover the kernel window. **Padding** solves this by adding extra values, often zeros, around the image border. With **same padding**, the output keeps the same spatial dimensions as the input. With **valid padding**, no extra border is added, so the output becomes smaller. Padding therefore affects both output size and how border regions are treated.

Some noteworthy architectures throughout the years...

LeNet introduced end-to-end CNN-based image recognition



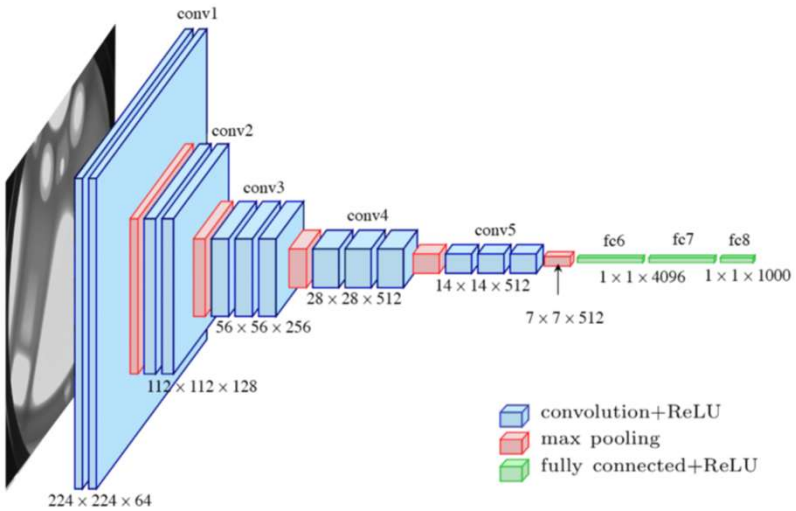
Y. LeCun, et al. Backpropagation Applied to Handwritten Zip Code Recognition. Neural Comput (1989) doi:10.1162/neco.1989.1.4.541

19

LeNet was one of the earliest successful convolutional neural networks and helped establish the basic CNN design: convolution, pooling, and fully connected classification layers. Its importance lies in showing that image-recognition features could be learned directly from data in an end-to-end manner, rather than being manually engineered in advance.

Some noteworthy architectures throughout the years…

VGG **showed that simple, deeper CNNs with small filters greatly improved accuracy**



K Simonyan & A Zisserman. Very Deep Convolutional Networks for Large-Scale Image Recognition. arXiv (2014) doi:10.48550/arXiv.1409.1556

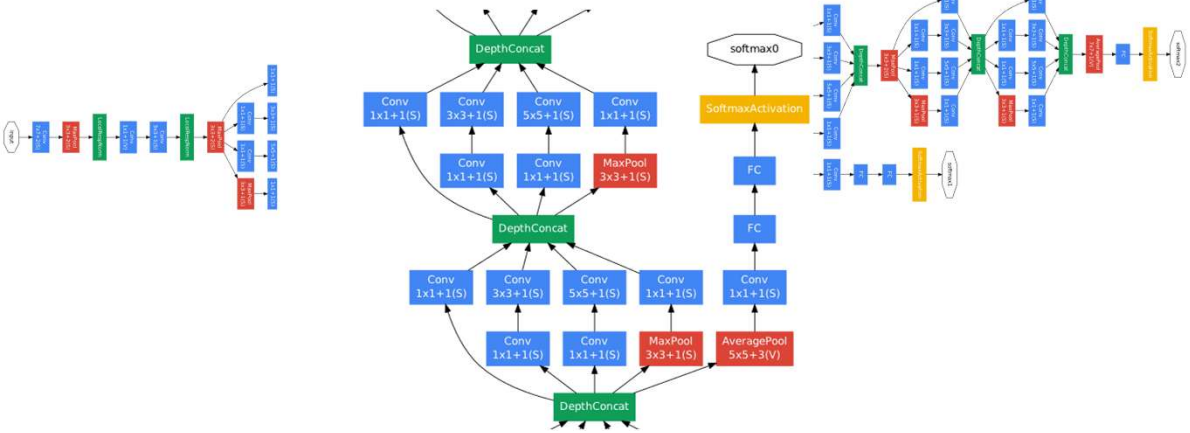
20

VGG showed that CNN performance could be improved by using a simple but much deeper architecture built from repeated small convolutional filters. Its design was influential because it demonstrated that depth itself was powerful, even without highly complicated architectural components. VGG also became a widely used backbone and feature extractor in many later computer-vision applications.

Note how the size of the feature maps decreases across layers (224 to 7), but the number of channels increases (64 to 512). This shows how the model is condensing spatial information into a non-spatial output.

Some noteworthy architectures throughout the years

Inception **introduced parallel convolutions with varying kernel sizes and intermediate output layers for improved learning**



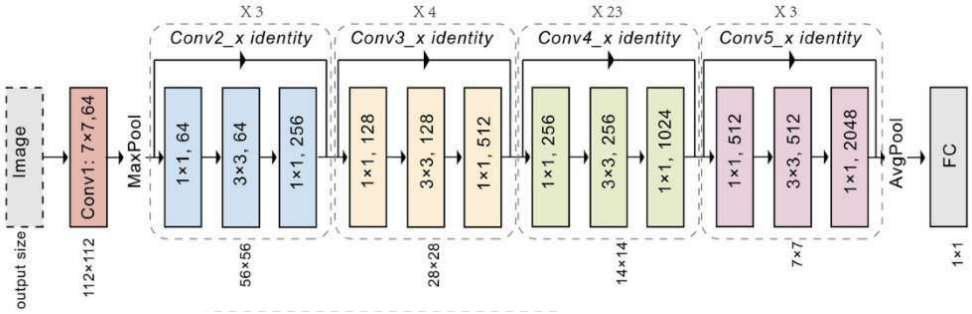
C Szegedy et al. Going Deeper with Convolutions. arXiv (2014) doi:10.48550/arXiv.1409.4842

21

The Inception architecture introduced modules that apply multiple convolutional operations in parallel, often with different kernel sizes. This allows the network to capture image patterns at multiple spatial scales within the same layer. The architecture was important because it improved both performance and computational efficiency, showing that CNN design could be made more sophisticated than simply stacking layers ever deeper.

Some noteworthy architectures throughout the years

ResNet made very deep CNNs trainable through residual skip connections



K He et al. Deep Residual Learning for Image Recognition. arXiv (2015) doi:10.48550/arXiv.1512.03385

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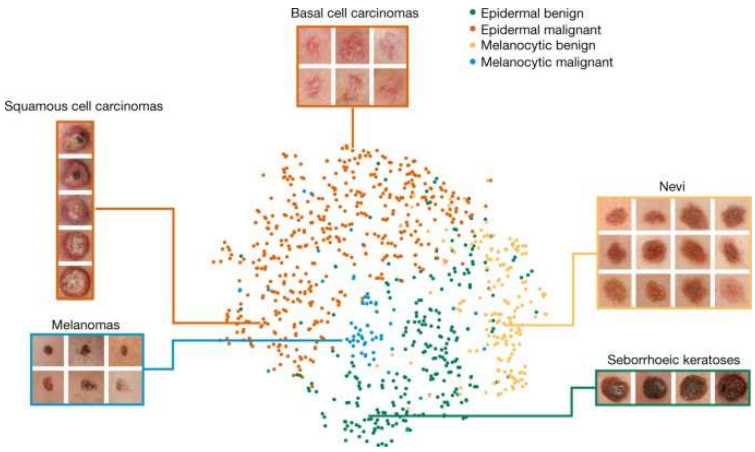
ResNet introduced residual skip connections, which allow information to bypass one or more layers. This made it much easier to train very deep networks by improving gradient flow during optimization. ResNet was a major milestone because it showed that substantially deeper CNNs could be trained effectively and could achieve much stronger performance on difficult visual-recognition tasks.

Our dataset for this afternoon: Skin lesions with CNNs

One of the first CNNs that reached expert-level performance



Esteva et al., Dermatologist-level classification of skin cancer with deep neural networks, Nature (2017)

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This example shows a landmark medical-imaging application of CNNs: skin-lesion classification. The study became highly influential because it suggested that deep neural networks could achieve expert-level performance on a clinically relevant image-recognition problem. More broadly, it helped establish deep learning as a serious approach in biomedical image analysis.

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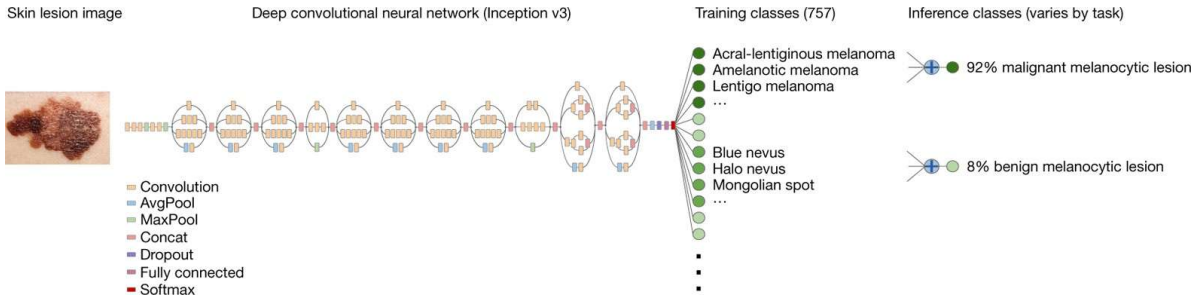

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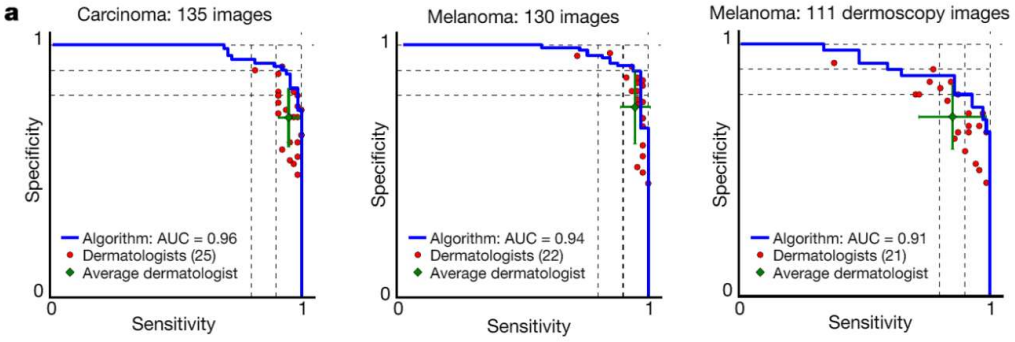

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